

Collin Cranston

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EDUCATION

University of California San Diego

PhD in Mathematics, Specialization in Statistics

M.S. in Statistics

San Diego, CA

Sep. 2020 – June 2026

Mar. 2023

- **PhD Thesis:** *Random Matrix Theory for Linearized Neural Networks*

- **Awards:** James B. Ax Fellowship

University of California, Los Angeles

B.S. in Mathematics, Highest Departmental Honors

Los Angeles, CA

Sep. 2015 – June 2019

PUBLICATIONS & PREPRINTS

NTK Regression under Dual Power-Law Model: Deterministic Equivalents via SDE and PDE Methods

C. Cranston, Z. Wang, T. Kemp

NeurIPS (Main), 2026

Eigen-Spike Emergence and Quadratic Equivalents for Conjugate Kernels on Nonlinearly Separable Data

C. Cranston, Z. Wang*, T. Kemp, M. Mahoney (*equal contribution)*

COLT, 2026

arXiv:2605.29669

EXPERIENCE

Data Science Intern

Cotality

June 2026 – Aug. 2026

San Diego, CA

- Built a prototype multi-agent system (Google ADK 2.0, Vertex AI RAG Engine) that drafts state department insurance filings for Cotality's predictive models, grounding each answer in prior filings.
- Designed a deterministic graph-based orchestrator with human-in-the-loop checkpoints, LLM-as-a-judge validation, and audit logging to meet compliance requirements.
- Evaluated on multiple historical filings, with projected filing time reduced from several weeks to just hours. Presented agentic workflow in multiple organization-wide sessions.

AI Trainer Mathematics Expert

Handshake AI

June 2025 – Aug. 2025

Remote

- Created and evaluated over 200 original tasks in topics ranging from complex analysis to random matrix theory that were used to assess and fine-tune performance of large language models (LLMs) in mathematical reasoning

Graduate Student Researcher & Teaching Assistant

University of California San Diego

Sep. 2020 – June 2026

San Diego, CA

- PhD thesis contributed to deep learning theory by deriving precise test error prediction as a function of training size, model width, and data spectrum; published two first/co-first author papers in A* conferences (NeurIPS/COLT).
- Teaching assistant for the 2023 SLMATH Summer Graduate School on Machine Learning. Designed and led problem-solving sessions to accompany lectures on multidimensional scaling and reinforcement learning.

PROJECTS

Deep Learning | *Computer Vision, NLP*

- Trained deep CNNs (U-Net, ResNet) for semantic segmentation on PASCAL VOC-2012 achieving 0.408 IoU.
- Fine-tuned transformer encoder (BERT) for scenario text classification on Amazon MASSIVE dataset via LoRA and contrastive learning, achieving 84.5% and 91.6% accuracy respectively.

NTK Theoretical Simulation | *GPU, PyTorch*

- Built batched-GPU simulation suite to validate theoretical predictions for NTK regression paper, chunking kernel matrix over network width to bound peak memory allocation.

TECHNICAL SKILLS

Languages: Python, MATLAB, SQL, R

Tools: Git, Google Cloud Platform (GCS, BigQuery, Vertex AI)

Libraries: PyTorch, pandas, NumPy, scikit-learn, Matplotlib

Misc: Google ADK, MCP, RAG